



TULSIRAMJI GAIKWAD-PATIL College of Engineering and Technology

Wardha Road, Nagpur - 441108

Accredited with NAAC A+ Grade

Approved by AICTE, New Delhi, Govt. of Maharashtra



(An Autonomous Institute Affiliated to RTM Nagpur University with NAAC A+ Grade)

Department of Information Technology

Session 2024-25(Odd Semester)

VISION	MISSION
To emerge as a learning hub and center of excellence in the domain of Information Technology	[M1] To impart quality technical education through effective teaching learning process. [M2] To provide a platform to address societal issues as well as challenges faced by IT industries. [M3] To foster a culture of research and impart innovative and entrepreneurial skills in the field of IT. [M4] To ensure overall development of students and staff by inculcating knowledge and professional ethics as a part of lifelong learning.

End Semester

Semester: V

Programme: B. Tech Information Technology

Course: Advanced Programming with Java

Time: 3 Hours

Max Marks: 60 Marks

Instructions to candidates:

1. All questions are compulsory in Group A
2. Solve any two sub-question out of three in Group B
3. Assume suitable data wherever necessary
4. All question carries marks as indicated

BITS33501	Course Outcome
CO1	To classify core Java fundamentals.
CO2	To analyze arrays, packages, and interfaces to develop modular and reusable Java applications.
CO3	To develop and manage multithreaded applications.
CO4	To utilize inheritance and polymorphism principles to create scalable Java applications.
CO5	To demonstrate knowledge of JDBC for database connectivity.

Group A				
Que. 1	Answer the following Question	CO	BTL	Marks
a.	Define class.	CO1	2	2
b.	What is interface?	CO2	2	2
c.	Define thread.	CO	2	2
d.	What is inheritance?	CO	3	2
e.	Define JDBC.	CO	2	2
Group B				
Que. 2	Answer the following Question (Solve any Two)	CO	BTL	Marks

a.	Explain JVM architecture.	CO1	2	5
b.	Explain data types and operators.	CO1	4	5
c.	Explain control structures.	CO1	3	5
Que 3.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Write program for array sorting.	CO2	2	5
b.	Explain packages and interfaces.	CO2	3	5
c.	Explain exception handling.	CO2	3	5
Que 4.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Write program for multithreading.	CO3	2	5
b.	Explain thread life cycle.	CO3	4	5
c.	Explain synchronization.	CO3	3	5
Que 5.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain inheritance with program.	CO4	4	5
b.	Explain overriding vs overloading.	CO4	3	5
c.	Explain types of inheritance.	CO4	3	5
Que 6.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain JDBC architecture.	CO5	2	5
b.	Explain driver types.	CO5	3	5
c.	Explain DB connectivity with example.	CO5	3	5

Bloom's Taxonomy Level (BTL)- Remember , Understand , Apply, Analyze , Evaluate , Create